# CBCS SCHEME

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BCS306A

## Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
## Object Oriented Programming with Java

Time: 3 hrs.   Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*
*2. M : Marks , L: Bloom's level , C: Course outcomes.*

| | | Module – 1 | M | L | C |
|---|---|---|---|---|---|
| Q.1 | a. | Discuss the different data types supported by Java along with the default values and literals. | 8 | L2 | CO1 |
| | b. | Develop a Java program to convert Celsius temperature to Fahrenheit. | 6 | L3 | CO2 |
| | c. | Justify the statement "Compile once and run anywhere" in Java. | 6 | L2 | CO1 |
| | | **OR** | | | |
| Q.2 | a. | List the various operators supported by Java. Illustrate the working of >> and >>> operators with an example. | 8 | L2 | CO1 |
| | b. | Develop a Java program to add two matrices using command line argument. | 10 | L3 | CO2 |
| | c. | Explain the syntax of declaration of 2D arrays in Java. | 2 | L2 | CO1 |
| | | **Module – 2** | | | |
| Q.3 | a. | Examine Java Garbage collection mechanism by classifying the 3 generations of Java heap. | 6 | L2 | CO1 |
| | b. | Develop a Java program to find area of rectangle, area of circle and area of triangle using method overloading concept. Call these methods from main method with suitable inputs. | 10 | L3 | CO2 |
| | c. | Interpret the general form of a class with example. | 4 | L2 | CO2 |
| | | **OR** | | | |
| Q.4 | a. | Outline the following keywords with an example :<br>(i) this<br>(ii) static | 6 | L2 | CO2 |
| | b. | Develop a Java program to create a class called 'Employee' which contains 'name', 'designation', 'empid' and 'basic salary' as instance variables and read ( ) and write ( ) as methods. Using this class, read and write five employee information from main ( ) method. | 10 | L3 | CO2 |
| | c. | Interpret with an example, types of constructions. | 4 | L2 | CO2 |
| | | **Module – 3** | | | |
| Q.5 | a. | Illustrate the usage of super keyword in Java with suitable example. Also explain the dynamic method dispatch. | 10 | L2 | CO3 |
| | b. | Build a Java program to create an interface Resizable with method resize (int radius) that allow an object to be resized. Create a class circle that implements resizable interface and implements the resize method. | 10 | L3 | CO3 |
| | | **OR** | | | |
| Q.6 | a. | Compare and contrast method overloading and method overriding with suitable example. | 8 | L2 | CO2 |

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|---|---|---|---|---|---|
| | b. | Define inheritance and list the different types of inheritance in Java. | 4 | L2 | CO3 |
| | c. | Build a Java program to create a class named 'Shape'. Create 3 sub classes namely circle, triangle and square ; each class has 2 methods named draw ( ) and erase ( ). Demonstrate polymorphism concepts by developing suitable methods and main program. | 8 | L3 | CO3 |
| | | **Module – 4** | | | |
| **Q.7** | **a.** | Examine the various levels of access protections available for packages and their implications with suitable examples. | **10** | **L2** | **CO4** |
| | **b.** | Build a Java program for a banking application to throw an exception, where a person tries to withdraw the amount even though he/she has lesser than minimum balance (Create a custom exception) | **10** | **L3** | **CO4** |
| | | **OR** | | | |
| **Q.8** | **a.** | Define Exception. Explain Exception handling mechanism provided in Java along with syntax and example. | **10** | **L2** | **CO4** |
| | **b.** | Build a Java program to create a package "balance" containing Account Class with displayBalance ( ) method and import this package in another program to access method of Account Class. | **10** | **L3** | **CO4** |
| | | **Module – 5** | | | |
| **Q.9** | **a.** | Define a thread. Also discuss the different ways of creating a thread. | **6** | **L2** | **CO5** |
| | **b.** | How synchronization can be achieved between threads in Java? Explain with an example. | **6** | **L2** | **CO5** |
| | **c.** | Develop a Java program for automatic conversion of wrapper class type into corresponding primitive type that demonstrates unboxing. | **8** | **L3** | **CO5** |
| | | **OR** | | | |
| **Q.10** | **a.** | Summarize the type wrappers supported in Java. | **6** | **L2** | **CO5** |
| | **b.** | Explain Autoboxing/Unboxing that occurs in expressions and operators. | **6** | **L2** | **CO5** |
| | **c.** | Develop a Java program to create a class myThread. Call the base class constructor in this class's constructor using super and start the thread. The run method of the class starts after this. It can be observed that both main thread and created child thread are executed concurrently. | **8** | **L3** | **CO5** |

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BCS306A

# Third Semester B.E./B.Tech. Degree Examination, June/July 2024
## Object Oriented Programming with Java

Time: 3 hrs. Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*
*2. M : Marks , L: Bloom's level , C: Course outcomes.*

| | | Module – 1 | M | L | C |
|---|---|---|---|---|---|
| Q.1 | a. | Differentiate two paradigms of programming. | 5 | L2 | CO1 |
| | b. | Explain the various bitwise and short circuit operators in Java. | 8 | L2 | CO1 |
| | c. | Write a Java program with a method to check whether a given number is prime or not. | 7 | L3 | CO1 |
| | | **OR** | | | |
| Q.2 | a. | Explain various scopes of variables in Java. | 5 | L2 | CO1 |
| | b. | Explain the arithmetic compound assignment and Ternary operators in Java. | 8 | L2 | CO1 |
| | c. | Write a Java program to perform linear search on an array elements accepted from keyboard and key element also accepted from key board. | 7 | L3 | CO1 |
| | | **Module – 2** | | | |
| Q.3 | a. | Explain method overloading in Java with examples. | 8 | L2 | CO2 |
| | b. | Design a stack class to hold maximum of N numbers with a constructor, push, POP and Display methods. Develop Java main method to illustrate stack operations. | 12 | L3 | CO2 |
| | | **OR** | | | |
| Q.4 | a. | Explain the role of "this" keyword and "static" keyword in Java. | 8 | L2 | CO2 |
| | b. | Design a class called "Employee" with fields ID, Name and Salary. Write a suitable constructors a method to raise salary and a static method to display. The number of Employee objects. Write suitable Main method for illustration. | 12 | L3 | CO2 |
| | | **Module – 3** | | | |
| Q.5 | a. | Explain the role of "Super" with example Java program. | 6 | L2 | CO3 |
| | b. | For any class and any method as an example, explain method overriding. | 5 | L2 | CO3 |
| | c. | Develop a Java program to create class called "Shape". Create 3 sub classes : circle, triangle and square. Each class has 2 member function area ( ) and draw ( ). Demonstrate polymorphism with a suitable main program. | 9 | L3 | CO3 |
| | | **OR** | | | |
| Q.6 | a. | Explain the order of constructor execution in a multilevel class hierarchy. | 6 | L2 | CO3 |
| | b. | Define dynamic method dispatch and write a code snippet in Java to demonstrate. | 5 | L1 | CO3 |

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|---|---|---|---|---|---|
| | c. | Develop Java program to create interface Resizable with methods resize width (int width) and resize height (int height) that allow object to be resized. Create a class Rectangle that implements This Interface. | 9 | L3 | CO3 |
| | | **Module – 4** | | | |
| **Q.7** | a. | Explain four categories of visibility for class members based on packages. | 6 | L2 | CO4 |
| | b. | Give the general form of an exception handling block and write a Java program to illustrate multiple catch classes. | 7 | L2 | CO4 |
| | c. | Write a custom exception in Java called "less marks" and raise This exception when marks entered by valuator in the range [30 – 34] | 7 | L3 | CO4 |
| | | **OR** | | | |
| **Q.8** | a. | With code snippets, explain mechanism to create and import a package in Java. | 6 | L2 | CO4 |
| | b. | Write a Java program to create chained exceptions with top-level exception is Null Pointer Exception and its cause Arithmetic Exception. | 7 | L3 | CO4 |
| | c. | Develop a Java program to create custom exception for Negative odd numbers. | 7 | L3 | CO4 |
| | | **Module – 5** | | | |
| **Q.9** | a. | Explain various methods of thread class in Java. | 6 | L2 | CO5 |
| | b. | Write a Java program to create 4 threads and each thread when run, will sleep for 500 milliseconds and print its name before "Before Quitting". | 8 | L3 | CO5 |
| | c. | Explain the use of Type wrappers in Java with example. | 6 | L2 | CO5 |
| | | **OR** | | | |
| **Q.10** | a. | Explain is Alive ( ) and join ( ) methods of Thread with example code snippet. | 6 | L2 | CO5 |
| | b. | Write a Java program to create 4 Rread and each Thread generates random number and prints it and sleeps for 800 msec and quits. | 8 | L3 | CO5 |
| | c. | Explain the concept of autoboxing /unboxing in expressions and methods. | 6 | L2 | CO5 |

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**Third Semester B.E./B.Tech. Degree Supplementary Examination, June/July 2024**

# Object Oriented Programming with Java

Time: 3 hrs. Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*
*2. M : Marks , L: Bloom's level , C: Course outcomes.*

| | | Module – 1 | M | L | C |
|---|---|---|---|---|---|
| Q.1 | a. | List and explain OOP's principles in JAVA. | 8 | L2 | CO1 |
| | b. | Class Helloworld {<br>Public static void main (String [ ] args) {<br>int a ;<br>for (a = 0 ; a < 3 ; a ++)<br>{<br>int b = -1 ;<br>system.out.println (" " +b) ;<br>b = 50 ;<br>system.out.println (" " +b) ;<br>}<br>system.out.println ("Hello, world!") ;<br>}<br>}<br>What is the output of the above code? | 6 | L3 | CO1 |
| | c. | Develop a program to find an average among the elements{1, 2, 3, 4, 5} using for each loop in JAVA. | 6 | L3 | CO1 |
| | | **OR** | | | |
| Q.2 | a. | How arrays are defined and used in Java? Give examples. | 6 | L2 | CO1 |
| | b. | Briefly explain the various primitive data types used in Java. | 6 | L2 | CO1 |
| | c. | Explain the following jump statements :<br>(i) Break (ii) Continue | 8 | L2 | CO1 |
| | | **Module – 2** | | | |
| Q.3 | a. | Explain the constructor method and parameterized constructors methods with suitable examples. | 10 | L3 | CO2 |
| | b. | Discuss the significant features of the following keyword :<br>(i) this<br>(ii) static | 4 | L2 | CO2 |
| | c. | What is method overloading? Illustrate the concept of method overloading using java program. | 6 | L2 | CO2 |
| | | **OR** | | | |
| Q.4 | a. | Write a java program to illustrate :<br>(i) Passing object as parameters.<br>(ii) Returning objects | 10 | L3 | CO2 |
| | b. | A class called Employee, which models an employee with an ID, name and salary. The method raiseSalary (percent) increases the salary by the given percentage. Develop the Employee Class and suitable main method for demonstration. | 10 | L3 | CO2 |

| | | | Marks | Level | CO |
|---|---|---|---|---|---|
| | | **Module – 3** | | | |
| Q.5 | a. | With example, give two uses of super. | 6 | L2 | CO3 |
| | b. | What is dynamic method dispatch? Write a simple example that illustrates dynamic method dispatch. | 8 | L2 | CO3 |
| | c. | Briefly explain the final keyword with inheritance. | 6 | L2 | CO3 |
| | | **OR** | | | |
| Q.6 | a. | What is an interface? Briefly explain the general forms of an interface. | 6 | L2 | CO3 |
| | b. | Discuss the significance of nested interfaces in Java. | 8 | L2 | CO3 |
| | c. | With proper syntax, explain the method overriding. | 6 | L2 | CO3 |
| | | **Module – 4** | | | |
| Q.7 | a. | What is a package? What are the steps involved in creating user defined packages? Explain. | 10 | L2 | CO4 |
| | b. | Define exception and explain the exception handling mechanism with an example. | 6 | L2 | CO4 |
| | c. | Discuss about throw and throws features. | 4 | L2 | CO4 |
| | | **OR** | | | |
| Q.8 | a. | Write a program to illustrate for nested try statements. | 6 | L2 | CO4 |
| | b. | Enlist any three java Built-in exceptions and explain. | 6 | L2 | CO4 |
| | c. | What is chained exception? Give an example that illustrates the mechanics of handling chained exceptions. | 8 | L2 | CO4 |
| | | **Module – 5** | | | |
| Q.9 | a. | Write a program to create multiple threads in JAVA. | 10 | L3 | CO5 |
| | b. | With syntax, explain the use of isAlive ( ) and join ( ) methods. | 6 | L3 | CO5 |
| | c. | Discuss the significance of thread priorities in JAVA. | 4 | L2 | CO5 |
| | | **OR** | | | |
| Q.10 | a. | With Syntax, explain values ( ) and value of ( ) methods. | 6 | L2 | CO4 |
| | b. | List and Discuss the Numeric type wrappers methods. | 6 | L2 | CO4 |
| | c. | Write a program to demonstrate the following :<br>(i) A type wrapper<br>(ii) Autoboxing/Unboxing | 8 | L2 | CO4 |

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# Model Question Paper-I/II with effect from 2023-24 (CBCS Scheme)

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## Third Semester B.E. Degree Examination

### Object Oriented Programming with JAVA

**TIME: 03 Hours** **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
02. Use a JAVA code snippet to illustrate a specific code design or a purpose.

| | | **Module -1** | ***Bloom's Taxonomy Level** | **Marks** |
|---|---|---|---|---|
| Q.01 | a | Explain different lexical issues in JAVA. | L2 | 7 |
| | b | Define Array. Write a Java program to implement the addition of two matrixes. | L3 | 7 |
| | c | Explain the following operations with examples. (i)<< (ii)>> (iii)>>> | L2 | 6 |
| | | OR | | |
| Q.02 | a | Explain object-oriented principles. | L2 | 7 |
| | b | Write a Java program to sort the elements using a for loop. | L3 | 7 |
| | c | Explain different types of if statements in JAVA | L2 | 6 |
| | | **Module-2** | | |
| Q. 03 | a | What are constructors? Explain two types of constructors with an example program. | L3 | 7 |
| | b | Define recursion. Write a recursive program to find nth Fibonacci number. | L3 | 7 |
| | c | Explain the various access specifiers in Java. | L2 | 6 |
| | | OR | | |
| Q.04 | a | Explain call by value and call by reference with an example program | L3 | 7 |
| | b | Write a program to perform Stack operations using proper class and Methods. | L3 | 7 |
| | c | Explain the use of **this** in JAVA with an example. | L2 | 6 |
| | | **Module-3** | | |
| Q. 05 | a | Write a Java program to implement multilevel inheritance with 3 levels of hierarchy. | L3 | 7 |
| | b | Explain how an interface is used to achieve multiple Inheritances in Java. | L3 | 7 |
| | c | Explain the method overriding with a suitable example. | L2 | 6 |
| | | OR | | |
| Q. 06 | a | What is single-level inheritance? Write a Java program to implement single-level inheritance. | L3 | 7 |
| | b | What is the importance of the **super** keyword in inheritance? Illustrate with a suitable example. | L3 | 7 |
| | c | What is abstract class and abstract method? Explain with an example. | L2 | 6 |
| | | **Module-4** | | |
| Q. 07 | a | Define package. Explain the steps involved in creating a user-defined package with an example. | L2 | 7 |
| | b | Write a program that contains one method that will throw an IllegalAccessException and use proper exception handles so that the exception should be printed. | L3 | 7 |
| | c | Define an exception. What are the key terms used in exception handling? Explain. | L2 | 6 |
| | | OR | | |
| Q. 08 | a | Explain the concept of importing packages in Java and provide an example demonstrating the usage of the import statement. | L2 | 7 |
| | b | How do you create your own exception class? Explain with a program. | L3 | 7 |
| | c | Demonstrate the working of a nested try block with an example. | L2 | 6 |

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|---|---|---|---|---|
| **Module-5** | | | | |
| Q. 09 | a | What do you mean by a thread? Explain the different ways of creating threads. | L2 | 7 |
| | b | What is the need of synchronization? Explain with an example how synchronization is implemented in JAVA. | L3 | 7 |
| | c | Discuss values() and value Of() methods in Enumerations with suitable examples. | L2 | 6 |
| **OR** | | | | |
| Q. 10 | a | What is multithreading? Write a program to create multiple threads in JAVA. | L2 | 7 |
| | b | Explain with an example how inter-thread communication is implemented in JAVA. | L3 | 7 |
| | c | Explain auto-boxing/unboxing in expressions. | L2 | 6 |